$$M_L \quad (V_L, U_L, R_L(V_L), F_L) \longrightarrow M_H \quad (V_H, U_H, R_H(V_H), F_H)$$

No "abstraction" $\Longrightarrow V_L = V_H$.

- approximate abstraction distance

d_τ = "worst case error" distance

d_{max}.

$\tau : R(V_L) \longrightarrow R(V_H)$ [state mapping]

$$d_\tau(M_L, M_H) = \min_{\{\tau_u : \tau_u(Pr_L) = Pr_H\}} \max_{\{\vec{X} \leftarrow \vec{x} \in I\}}$$

Expectation

$$\left(\sum_{\vec{u}_L \in R_L(U_L)} Pr_L(\vec{u}_L)\, d_{V_H}\Big(\underbrace{\tau(M_L(\vec{u}_L, \vec{X} \leftarrow \vec{x})), M_H(\tau_u(\vec{u}_L); \omega_\tau(\vec{X} \leftarrow \vec{x}))}_{\text{Difference}} \Big) \right)$$

$$\tau\left(Pr_L^{\vec{X} \leftarrow \vec{x}}\right) = Pr_H^{\omega_\tau(\vec{X} \leftarrow \vec{x})} \checkmark$$

$\min \tau_u \longrightarrow$ best case

$\max \vec{X} \leftarrow \vec{x} \longrightarrow$ worst case Intervention.

$\tau : R(V_L) \longrightarrow R(V_H)$

Intervention distrib. Pr_I is a distrib. on $I_L \times I_H$ s.t

$$Pr_I(\vec{X} \leftarrow \vec{x} \mid \vec{Y} \leftarrow \vec{y}) > 0 \text{ iff}$$

$$\underbrace{\omega_\tau(\vec{X} \leftarrow \vec{x})}_{\text{lifting this low level intervention}} = \underbrace{\vec{Y} \leftarrow \vec{y}}_{\text{gives us this high level one}}$$

$$d_\tau^{Pr_I}(M_L, M_H) = \min_{\{\tau_u : \tau_u(Pr_L) = Pr_H\}} \max_{\vec{Y} \leftarrow \vec{y} \in I_H}$$

$$\Bigg[\underbrace{\sum_{\vec{u}_L \in R_L(\vec{U}_L)} Pr_L(\vec{u}_L)}_{1} \; \underbrace{\sum_{\vec{X} \leftarrow \vec{x} \in I_L} Pr_I(\vec{X} \leftarrow \vec{x} \mid \vec{Y} \leftarrow \vec{y})}_{2}$$

$$\underbrace{d_{V_H}\big(\tau(M_L(\vec{u}, \vec{X} \leftarrow \vec{x})), M_H(\tau_u(\vec{u}), \omega_\tau(\vec{X} \leftarrow \vec{x}))\big)}_{\text{unchanged, same formula}} \Bigg]$$

1 averages error accross all likely low level contexts? weighted by probab. Pr_L

2 average error accross all possible low level implementations $(X \leftarrow x)$ of that high level action, weighted by the interventional distrib. Pr_I.

①

$$\begin{tikzcd}
(M_H, I_H) \arrow[r, "d_{max}\,\alpha \text{ approximation}"] & (M'_H, I_H) \\
(M_L, I_L) \arrow[u, "\tau\text{-abstraction}"'] &
\end{tikzcd}$$

$\Longrightarrow (M'_H, I_H)$ is an τ-α approximate abstraction of (M_L, I_L)

② Opposite order:

$$\begin{tikzcd}
\boxed{M'_L} \arrow[rr] & & dV_H \\
\boxed{M_L} \arrow[u, "d_{max}\,\alpha \text{ approximation}"'] \arrow[r, "\tau\text{-abstraction}"'] \arrow[d] & \boxed{M_H} \arrow[ur] & \\
dV_L & &
\end{tikzcd}$$

$\Longrightarrow$ we can't say that M_H is a $d_\tau\,\alpha$ approxim. of M_L

Thereom 5.3:

*(M_H is noisy version of M'_H)

$$\begin{tikzcd}
(M_H, I_H) \arrow[dr, bend right, "\text{does this exist?}"'] & (M'_H, I_H) \\
(M_L, I_L) \arrow[u, "\tau\text{-}\alpha \text{ approximate abstraction}"] \arrow[ur, "\text{does this exist?}"'] & (M'_L, I_L)
\end{tikzcd}$$

(M'_H, I_H): A perfect high level model (exact abstraction)

(M'_L, I_L): perfect low level model that M_H abstracts exactly